1. We can split this up into two cases:

   Case 1: n is a perfect square
   In this case, $\sqrt{n}$ is an integer. If it is prime, then n has a prime factor less than or equal to $\sqrt{n}$. If not, then by the Fundamental Theorem of Arithmetic, $\sqrt{n}$ has a prime factorization, and therefore, n has prime factors less than or equal to $\sqrt{n}$.

   Case 2: n is not a perfect square
   We know that n = ab for positive integers a and b, since n is composite. If a and b were both greater than $\sqrt{n}$, ab would be greater than n, which we know can't be true. That means at least one of them is less than or equal to $\sqrt{n}$. □

2. Let's assume there are finitely many primes of the form 6n + 5, which we can call p_1, p_2, …, p_n. Now, let's consider $k = \prod_{i=1}^{n} p_i - 1$. Also, every prime must be either of the form 6n + 1 or 6n + 5. From here, we can proceed with two cases:

   Case 1: k is prime
   In this case, we have found a prime of the form 6n - 1 = 6(n - 1) + 5 that isn't one of p_i. This is a contradiction.

   Case 2: k is composite
   In this case, it must be divisible by a number of the form 6n + 1 or 6n + 5. Multiplying two numbers of the form 6n + 1 gives us (6x + 1)(6y + 1) = 36xy + 6x + 6y + 1 = 6(6xy + x + y) + 1, which is also of the form 6n + 1. That means, if k is divisible by a prime of the form 6n + 1, k is of the form 6n + 1 as well, but clearly, it can't be. That means it is divisible by a prime of the form 6n + 5. Since it's not divisible by any of p_i, we have found a prime that is not on our finite list of primes of that form. So we have a contradiction, and there are infinitely many primes of the form 6n + 5. □

3. Let's say two factors of n are x and y, where xy = n. Also, let's say that a = cx and b = dy where c<x and d<y. That means ab = cdxy, which is a multiple of n. But also, neither a nor b are multiples of n because cx < n and dx < n. □

4. For the sake of contradiction, let's assume that none of a_1, a_2,... a_n are multiples of p. Let's say the product of those numbers is $\prod_{i=1}^{k} {q_k}^{b_i}$ for prime numbers q_1, q_2,... q_n, none of which are p, a list of exponents b, and some positive integer k. That means the product $\prod_{i=1}^{n} a_i$ is written as the product of primes raised to some powers, none of which are p.
   That means p is not a factor of the product, which is a contradiction to the statement in the problem that p is a factor of the product. Therefore, our original assumption that none

of a_1, a_2,... a_n are multiples of p is wrong, and at least one of them must be a multiple of p. □

5. For integers that satisfy gcd(a, b, c) = 1, there exist integers x, y, and z such that ax + by + cz = 1.

   If gcd(a, b) = 1, there exist integers x_0 and y_0 such that $ax_0 + by_0 = 1$. Now, let's say gcd(a, b, c) = 1. This means that gcd(c, $ax_0 + by_0$) = 1, because gcd(gcd(a, b), c) = gcd(a, b, c). That means there exist integers w_0 and z_0 such that $cz_0 + (ax_0 + by_0)w_0 = 1$. Expanding that, we get $ax_0w_0 + by_0w_0 + cz_0 = 1$. This means, if $x = x_0w_0$, $y = y_0w_0$, and $z = z_0$, then ax + by + cz = 1. □

6. The integers 5, 8, 10, 13, 15, 16, 18, 20, 21, 23, 24, 25, 26, 28, 29, 30, and every integer greater than 30 can be expressed in the form 5x + 8y for nonnegative integers x and y. To prove this, we can create a table:

0	1	2	3	4	5	6	7	8	9
10	11	12	13	14	15	16	17	18	19
20	21	22	23	24	25	26	27	28	29
30	31	32	33	34	35	36	37	38	39

   Then, we can change the color of numbers that can be expressed as 5x + 8y to blue.

0	1	2	3	4	5	6	7	8	9
10	11	12	13	14	15	16	17	18	19
20	21	22	23	24	25	26	27	28	29
30	31	32	33	34	35	36	37	38	39

   We see that every number in the 30s can be expressed as 5x + 8y. This means, to find x and y for any number greater, we just need to find the units digit of that number and match it to the number in the 30s with the same units digit. Then, adding some specific even multiple of 5 to that number will result in the number we are creating. □

7. From the given equation, we can isolate b_2 to get $b_2 = \frac{a_2b_1 \pm 1}{a_1}$. Substituting this into $\frac{a_1+a_2}{b_1+b_2}$, we get $\frac{a_1+a_2}{b_1+b_2} = \frac{a_1+a_2}{b_1+\frac{a_2b_1\pm1}{a_1}} = \frac{a_1(a_1+a_2)}{b_1(a_1+a_2)\pm1}$. We know that gcd(an, bn ± 1) = 1, because gcd(an, bn) is at least n, and adding or subtracting 1 would make them relatively prime.

That means the numerator and denominator are relatively prime, and so $\frac{a_1+a_2}{b_1+b_2}$ cannot be simplified. □

8. We can prove this to be true by showing that if f(n) is prime for all integers n, then f(x) is constant. First, we note that if f(x) is prime, then f(0) = p for some prime p. If p wasn't prime, you could plug in n = p and f(n) would be a multiple of p. Also, we know that f(pn) is a multiple of p, because you can factor out p from $a_n(pn)^n + a_{n-1}(pn)^{n-1} + \ldots + a_1(pn) + p$. Since p divides f(pn) but f(pn) is prime, f(pn) = p for all n. Subtracting p, we get f(pn) - p = 0, and all integers n are solutions. That means f(px) - p has infinitely many solutions, and so it is equal to 0. That means f(x) is constant, which is a contradiction to the statement in the problem that f(x) is nonconstant. That means there exists n such that |f(n)| is composite. □

9. Here are the proofs:
    a. Let's prove this through induction. We can start with the base case, which is $\gcd(F_0, F_1) = 1$. We see that is true, because gcd(0, 1) = 1.

       Now, for the inductive step, let's assume $\gcd(F_n, F_{n+1}) = 1$. To find $\gcd(F_{n+1}, F_{n+2})$, we can use the Euclidean Algorithm. We see that is equal to $\gcd(F_{n+1}, F_{n+2} - F_{n+1}) = \gcd(F_{n+1}, F_n)$, which we assumed was 1. Therefore, $\gcd(F_{n+1}, F_{n+2}) = 1$. □

    b. I tried proving this through induction, starting with the base case: $\gcd(F_1, F_n) = 1 = F_{\gcd(1, n)}$.

       Then, for the inductive step, we can assume $\gcd(F_m, F_n) = F_{\gcd(m, n)}$ and prove that $\gcd(F_{m+1}, F_n) = F_{\gcd(m+1, n)}$. But I didn't know how to proceed from there.

10. To prove this, we can prove the contrapositive, that if n is composite, then $2^n - 1$ is not prime.

    Since n is composite, we can write it as ab where a and b are positive integers greater than 1.
    We know that $x^b - 1 = (x - 1)(x^{b-1} + x^{b-2} + \ldots + x + 1)$. Replacing x with 2^a, we get $2^n - 1 = 2^{ab} - 1 = (2^a - 1)((2^a)^{b-1} + (2^a)^{b-2} + \ldots + 2^a + 1)$. We just wrote $2^n - 1$ in terms of two integers greater than one, so we see that $2^n - 1$ is composite. Since a statement and its contrapositive are logically equivalent, we have proven the original statement as well. □

11. If n is odd, we know that $x^n + 1 = (x + 1)(x^{n-1} - x^{n-2} + \ldots + 1)$. If m = nd, then $x^d + 1$ divides $x^m + 1$, because of the factorization $x^m + 1 = (x^d)^n + 1 = (x^d + 1)((x^d)^{n-1} - \ldots + 1)$. That means $2^m + 1$ is divisible by $2^d + 1$, so it's not prime. So if it is prime, then m cannot have an odd factor, which means it is a power of 2. □